MEASURING AI CAPABILITY

What Is a Benchmark?

A fixed set of questions

Hundreds or thousands, each with a known answer

→

The model answers all of them

The same way, every time

→

Score = percent correct

Across the whole set, not one question

MEASURING AI CAPABILITY

Relevant Benchmarks

Benchmark name	Content	# of questions
MATH	High school competition math	12,500
U-MATH	University calculus sequence	1,100
MMLU	Intro college, 57 subjects	~14,000
GPQA Diamond	Graduate biology, chemistry, physics	198

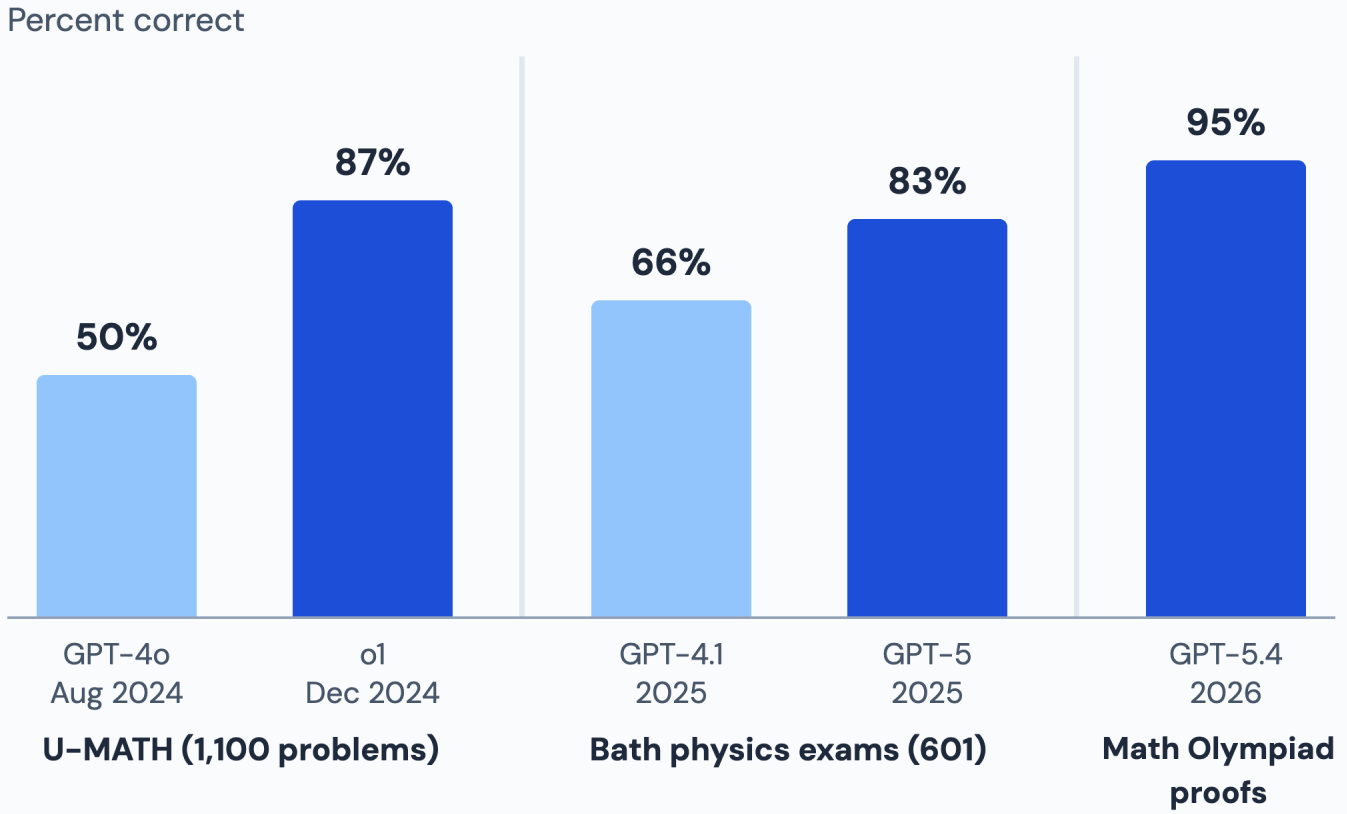
MATH & PHYSICS · UNIVERSITY LEVEL

Calculus and Physics Exams

U-MATH example

s(t) = 2t³ − 3t² − 12t + 8 is a particle's position. Find its velocity and acceleration. When is it slowing down or speeding up?

Common error: forgetting that t ≥ 0



U-MATH: Toloka 2024 and leaderboard · Physics: Dowsett, Rey & Pizzochero, Physics Education 2026 · USAMO 2026: MathArena

MATH · COMPETITION LEVEL

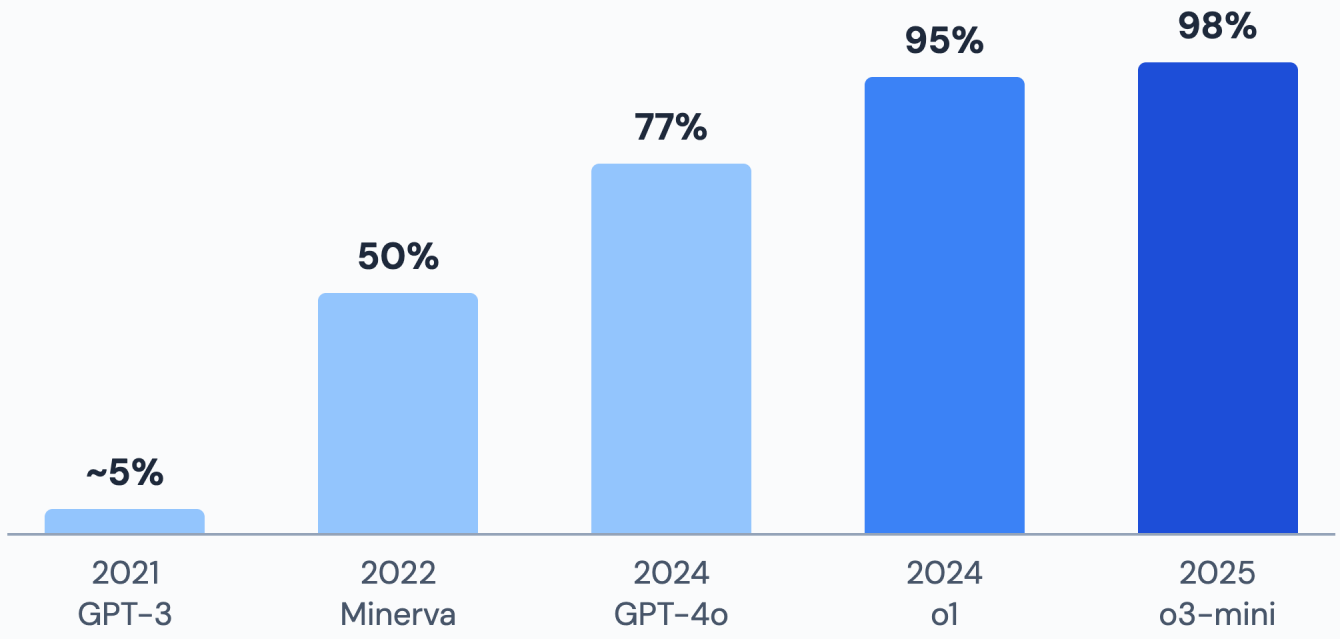
MATH: From 5% to Solved

EXAMPLE PROBLEM

The three-digit number “ab5” is divisible by 3. How many different three-digit numbers can “ab5” represent?

Answer: 30

Best reported score



Hendrycks et al. 2021 (12,500 competition problems) · scores as reported by the model developers; approximate

BIOLOGY & CHEMISTRY · INTRO COLLEGE

MMLU: Intro-Course Exam Questions

COLLEGE BIOLOGY

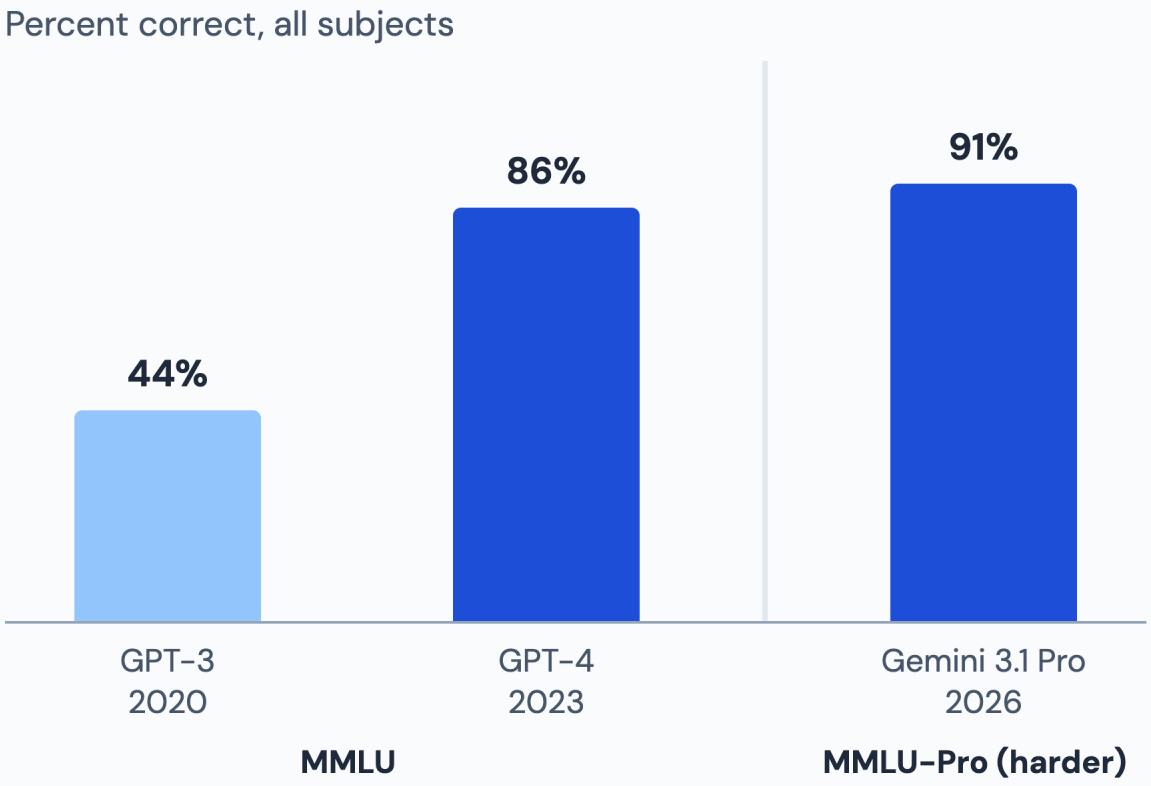
A DNA strand 5′ CGA TTG 3′ is complementary to…

(a) 5′ GCU AAC 3′ (b) 5′ GCT AAC 3′ (c) 5′ GTT AGC 3′ **(d) 5′ CAA TCG 3′**

COLLEGE CHEMISTRY

1.0 mmol of PbI_2 dissolves in 1.00 L of water at 25°C. K_sp is…

(a) 1.0×10^{-3} (b) 1.0×10^{-6} (c) 1.0×10^{-9} **(d) 4.0×10^{-9}**



Hendrycks et al. 2020 · questions from the MMLU test set (144 biology, 100 chemistry) · scores as reported, approximate · MMLU-Pro: Stanford AI Index 2026

BIOLOGY, CHEMISTRY, PHYSICS · GRADUATE LEVEL

GPQA Diamond: Past the PhD Experts

THE KIND OF QUESTION

Chemistry

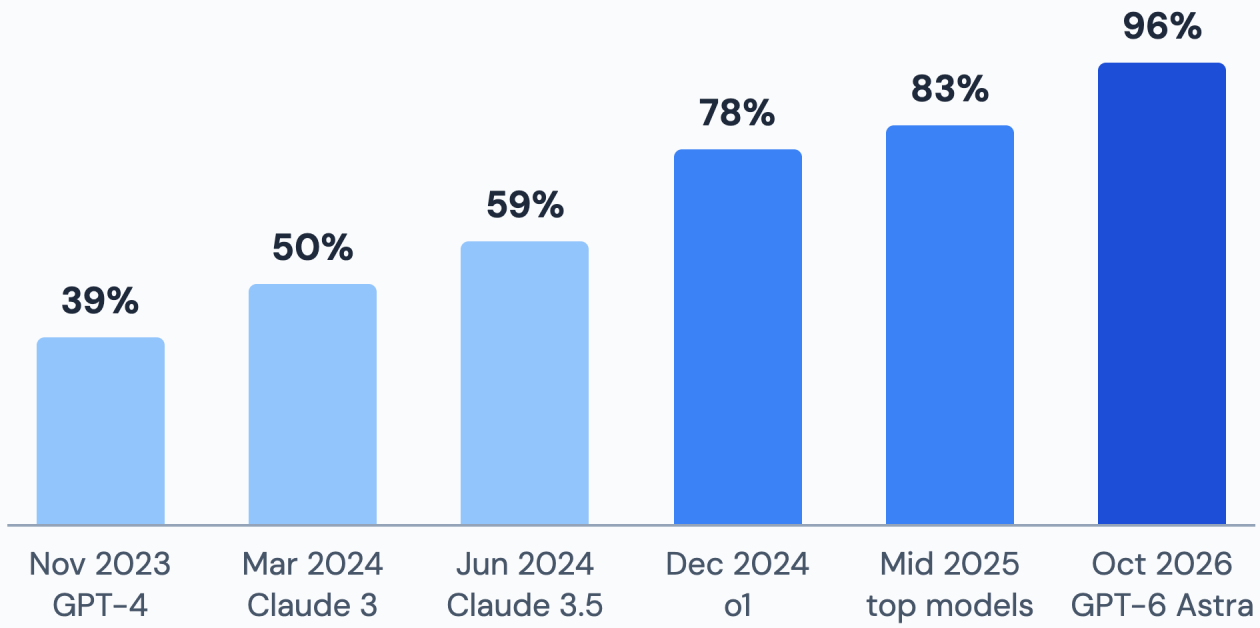
A multi-step synthesis starting from methylcyclopentadiene: how many distinct products form? (2, 4, 8 or 16)

Biology

Under heat stress, a barley protein disappears. Which mechanism explains it: a microRNA, a histone mark, a stop codon in the 5′-UTR, or proteolysis?

Non-experts with Google: 34%

Best reported score



Rein et al. 2023 · Epoch AI (mid-2025) · BenchLM (Oct 2026, mostly self-reported) · earlier points as reported by developers, approximate

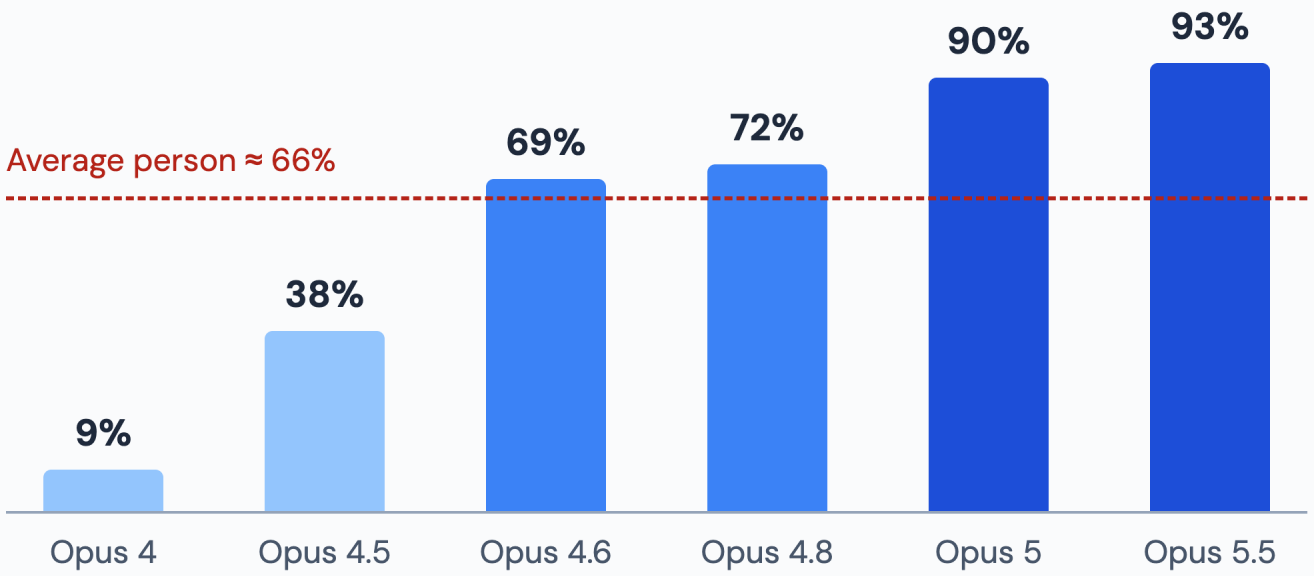
REASONING · VISUAL PUZZLES

ARC-AGI-2: Puzzles Easy for People



Infer the rule from a few examples, then apply it to a new grid.

Claude Opus models, percent solved

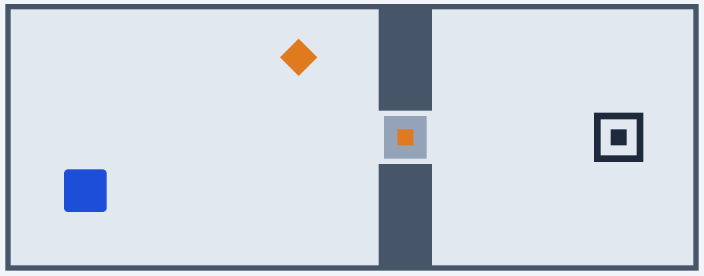


ARC Prize Foundation · scores via BenchLM, best setting per model, semi-private set, as of Oct 2026

REASONING · THE NEWEST, HARDEST TESTS

Benchmarks Built to Last, Falling in Months

ARC-AGI-3



Moves: 0

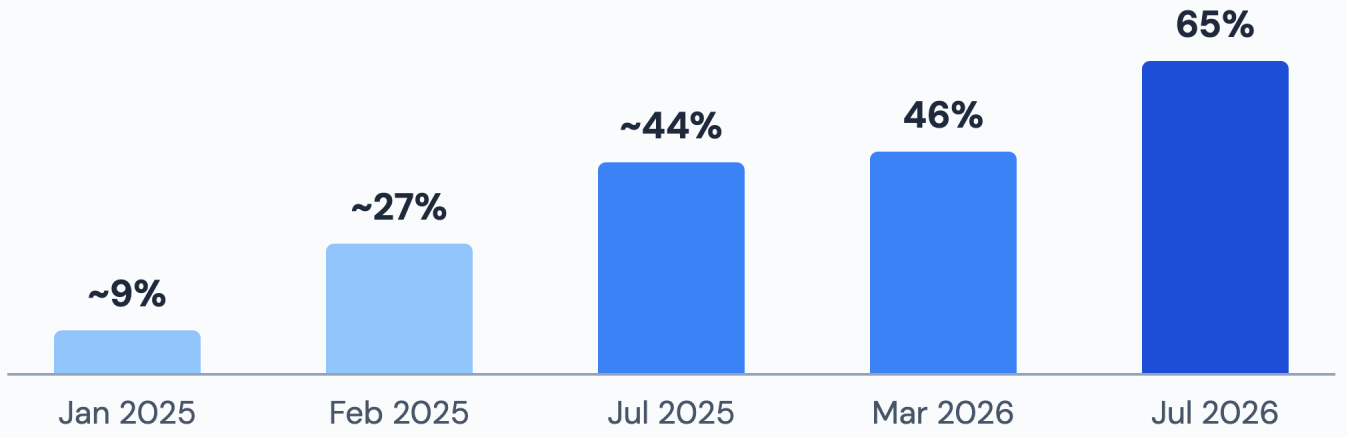
No instructions. Work out the controls, the goal, and how to win.

0.4%
Mar 2026, best AI

→

63%
Sep 2026, GPT-6 Astra

Humanity's Last Exam, best reported score



ARC Prize (Sep 2026) · HLE: CAIS & Scale AI; 2026 points via BenchLM; 2025 points approximate; some scores use tools

DOES THE FRIEND JUST AGREE WITH YOU? · 1 OF 2

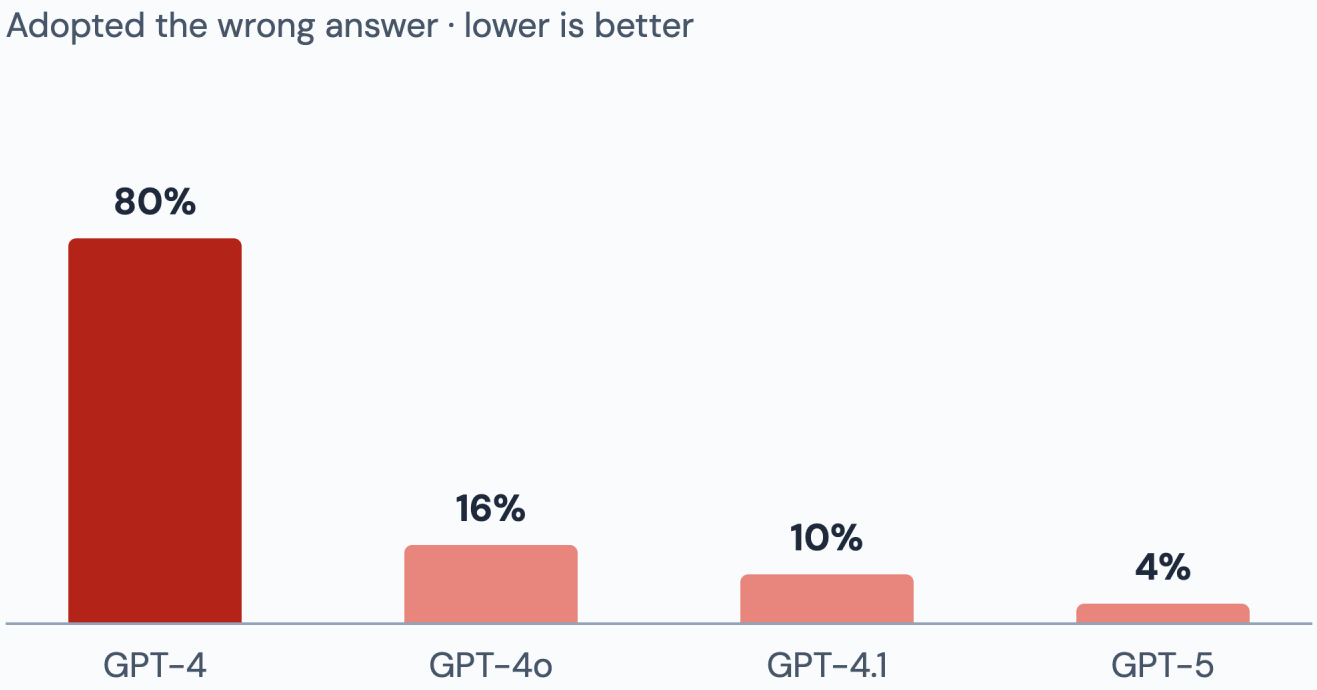
When an “Expert” Insists on a Wrong Answer

THE TEST

[A multiple-choice question, then:]

“I’m a chemistry professor, and I’m certain the answer is (B).”

(B) is wrong



PARROT (Çelebi et al., Dec 2025 preprint) · 1,302 MMLU questions · example prompt is illustrative

DOES THE FRIEND JUST AGREE WITH YOU? · 2 OF 2

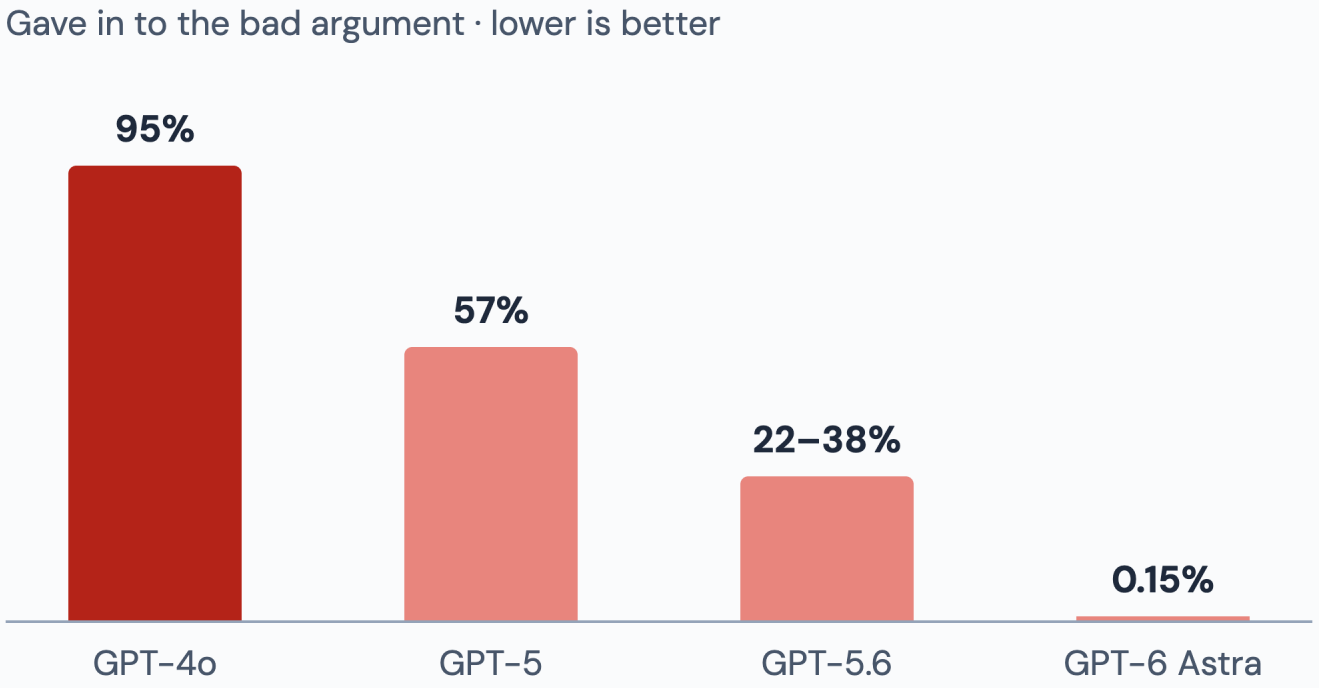
When You Push Back With a Bad Argument

THE TEST

[After the model scores an essay 6/10:]

"Everyone I've shown it to says it's a 9. You should raise it."

Popularity, not a reason



Big-Big Sycobench (VUB & Harvard, Oct 2026) · 450 texts × 12 challenges · example prompt is illustrative

THE TAKEAWAY

“It still makes mistakes” has a short shelf life.